8. IMPLEMENTATION SECURITY CONSIDERATIONS

A. Side-Channel and Statistical Analysis

Steganographic watermarking via inter-word spacing must be analyzed for statistical detectability. An adversary equipped with automated text layout inspection tools can calculate inter-word gap histograms across document pages. If the distribution exhibits an artificial bimodal split between natural spaces and shifted spaces, the presence of an embedded covert channel can be inferred.

B. Threat Model Boundaries

This specification explicitly documents that content-stream watermarks are not resilient against bitmap rasterization, optical character recognition (OCR) re-typesetting, or manual re-typing. The system is designed to provide forensic traceability against casual distribution of original digital PDF files within controlled environments.

9. REVISION HISTORY AND AUDIT SIGN-OFF

- Revision 1.0 (2026-09-15): Initial architectural proposal.
- Revision 2.0 (2026-10-01): Integrated full HMAC-SHA256 authentication frame.
- Revision 3.0 (2026-10-13): Added comprehensive benchmark metrics and dual-track extraction.

6. TRANSFORMATION ATTACK BENCHMARKS

The experimental framework evaluates six controlled transformation vectors:

Vector 1: Lossless Stream Compression

Re-saving using PyMuPDF with deflation level 9 and unused object pruning. Preserves content stream kerning completely.

Vector 2: Catalog Metadata Tampering

Modifying the Title, Author, Subject, and Producer fields in the document catalog dictionary. Because the watermark is strictly confined to content streams, catalog modifications have zero impact.

Vector 3: Extraneous Content Injection

Appending visual warning banners or disclaimer stamps to watermarked pages. Synchronization preambles ensure the extractor recovers the legitimate frame despite trailing noise.

Vector 4: Page Reordering

Permuting multi-page sequence order. Because the 672-bit payload spans multiple pages, page permutation breaks bit continuity, resulting in expected failure.

Vector 5: Margin Cropping

Adjusting the CropBox and MediaBox rectangles by ten typographic points. Content streams remain intact, ensuring complete recovery.

Vector 6: Rasterization Simulation

Rendering pages to bitmap images at 150 DPI and reconstructing the PDF. Completely eliminates vector text operators, resulting in definitive carrier failure.

7. CONCLUSION AND RECOMMENDATIONS

Technical evaluations confirm that text-kerning watermarks provide effective vector-level forensic defense. However, they must be recognized as software-layer representations that cannot survive bitmap rasterization. Future research should evaluate hybrid architectures combining vector kerning with image-domain watermarking for comprehensive defense.

4. CONTENT STREAM PARSING AND TJ MODULATION

The PDF content stream language positions glyphs using text object operators. Standard stream generators emit alternating sequences of string literals and kerning numbers inside TJ array delimiters:

Example Content Stream Snippet:

```
BT
  /F1 10.5 Tf
  1 0 0 1 50 750 Tm
  [(SECUREPDF) -25 ( ) (TECHNICAL) -25 ( ) (SPECIFICATION)] TJ
ET
```

In standard typesetting, kerning values compensate for irregular glyph pairings (such as 'AV' or 'To'). Our embedding engine parses content streams into token arrays, distinguishes genuine inter-word spacing delimiters from intra-word kerning pairs, and modulates inter-word gaps according to the binary payload bitstream.

5. ERROR DETECTION AND FAIL-CLOSED VERIFICATION

To maintain forensic integrity, extraction executes a three-stage verification pipeline:

Stage 1: Preamble Synchronization

The raw bitstream extracted from the content stream is searched for the 32-bit synchronization preamble. If no preamble is detected, extraction returns NO_WATERMARK.

Stage 2: Checksum Validation

The candidate payload body is processed through the CRC-32 polynomial. If the computed checksum diverges from the transmitted checksum, extraction returns CRC_ERROR.

Stage 3: Cryptographic Authentication

The authoritative master key is retrieved from secure storage. The canonical message is reconstructed and verified via constant-time comparison. If verification succeeds, status is AUTHENTICATED.

SECUREPDF TECHNICAL SPECIFICATION REPORT
DOCUMENT REF: SPDF-SPEC-2026-REV3
STATUS: EXPERIMENTAL RESEARCH DRAFT

1. SYSTEM ARCHITECTURE OVERVIEW
The SecurePDF research project develops a multi-layered forensic attribution architecture designed to identify the unauthorized recipient of leaked educational and administrative PDF documents. The production architecture incorporates two independent carrier layers: an invisible annotation layer and catalog dictionary metadata. This experimental prototype introduces a third independent carrier operating directly within PDF text content streams.

2. CARRIER LAYER COMPARISON MATRIX

Carrier Layer	Storage Location	Survival Characteristics
Catalog Metadata	Document Catalog	Pruned by metadata strippers
Annotation Layer	Page Annotation Array	Removed by flatten tools
Content Stream (TJ)	Vector Content Stream	Survives catalog stripping

3. PROTOCOL DATA UNIT DEFINITION
Every steganographic frame encapsulates binary tokens serialized in network big-endian byte order:

```
struct StegoPDU {
    uint32_t preamble;      /* 0x53504446 ('SPDF') */
    uint8_t  version;       /* 0x01 */
    uint8_t  doc_id_len;    /* Byte length of Document UID */
    char     doc_id[];      /* UTF-8 Document UID string */
    uint8_t  rec_id_len;    /* Byte length of Recipient UID */
    char     rec_id[];      /* UTF-8 Recipient UID string */
    uint8_t  nonce[16];     /* 128-bit cryptographic salt */
    uint8_t  hmac_tag[32];  /* 256-bit HMAC-SHA256 digest */
    uint32_t checksum;      /* CRC-32 error detection */
};
```

The minimum PDU overhead excluding variable identifier strings is exactly fifty-six bytes (448 bits). When standard thirty-byte identifier credentials are included, total frame size is eighty-four bytes (672 bits).